

Ryan Khondker

Lead iOS Developer | Mobile Development | Cross-Platform (React Native; Kotlin)

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iOS Mobile Engineer with **13 years** of **IT** experience, including **11 years** of **iOS development** experience, for enterprise and consumer domains such as e-commerce, finance, healthcare, retail, and energy. Successfully published **6 iOS apps**, for Customer Facing Enterprise, for iPhone/iPad, on the App Store.

- **iOS Development Expertise:** Proven track record of delivering customer-facing iOS apps published on the App Store, leveraging Swift, Objective-C, SwiftUI, Combine, and Core frameworks to create reliable, scalable, and high-performance mobile solutions. Skilled in end-to-end iOS app lifecycle—from architecture, development, and integration to testing, deployment, and App Store submission.
- **Specialized in Fintech-Grade Security:** Delivered mobile banking and payment apps with strong authentication, encrypted data flows, and fraud prevention workflows. Applies privacy-by-design principles to meet compliance standards and build trustworthy user journeys.
- **Cross-Platform Exposure:** Proven experience in React Native, Flutter, and Kotlin Multiplatform (KMM) for building cross-platform mobile solutions alongside native iOS apps. Skilled at delivering scalable, performant, and reusable codebases across iOS and Android, accelerating time-to-market while maintaining native-level UX.
- **UI-UX & Mobile Architecture:** Adept at optimizing UI/UX with Apple HIG compliance, implementing modular architectures (MVC, MVVM, VIPER, Clean Architecture), and driving adoption of modern Apple technologies, including Core ML, ARKit, RealityKit, SiriKit, and visionOS APIs.
- **Leadership & Mentorship:** Recognized for mentoring teams, leading Agile/Scrum projects, and collaborating effectively with product owners, designers, and cross-functional engineering groups to ship cohesive, secure, and user-centric products. Strong focus on performance optimization, mobile security, and accessibility compliance, ensuring seamless experiences across iPhone, iPad, Apple Watch, and tvOS platforms.
- **Release, Operations, and reliability:** Manages TestFlight and App Store pipelines with phased rollouts, versioning, and certificate handling. Designs mobile solutions that handle poor networks, offline states, and device constraints with graceful degradation. Proactively monitors and resolves performance issues to maintain user trust and app ratings.
- **Global Reach & Data-Driven Product:** Experienced in localization, right-to-left layouts, and compliance with region-specific policies. Designs apps with flexible formats, currency handling, and global UX consistency for worldwide markets. Applies analytics and experimentation to refine user flows and optimize engagement.
- **Innovation & Growth:** Passionate about exploring next-gen Apple platforms, including visionOS, App Intents, and AI/ML integrations (LLM, ChatGPT). Maintains a continuous learning mindset, pushing innovation while ensuring apps are future-ready and maintainable.

TECHNICAL SKILLS

- **Programming Languages:** Swift, Objective-C, Kotlin, Java, C, C++, C#, .NET, Python, R, Scala, SQL, JavaScript, PHP, HTML, CSS, XML, Rust
- **Frameworks & Libraries:** SwiftUI, UIKit, Combine, RxSwift, ReactiveCocoa, Cocoa Touch, CocoaPods, Core Data, Core Animation, Core Graphics, Core Bluetooth, Core Location, EventKit, MapKit, RealityKit, ARKit, SceneKit, SpriteKit, AVFoundation, HealthKit, HomeKit, WatchKit, SiriKit, TipKit, StoreKit, Wallet & Apple Pay, Notification Frameworks (Local & Push), App Intents, Smart Stack, Double Tap API, Game Porting Toolkit 2, Metal, SF Symbols, Accessibility APIs, visionOS Volumetric APIs, TabletopKit, tvOS Enhancements, iPadOS Enhancements
- **Architecture & Design Patterns:** MVC, MVP, MVVM, VIPER, Clean Architecture, Dependency Injection, Modularization, OOP/OOD, Builder, Abstract Factory, Visitor, Singleton, Key Value Observing, Observer, Decorator, Delegation, Proxy, Interpreter, Iterator, Mediator
- **Mobile Development (iOS & Cross-Platform):** React Native, Flutter, Kotlin Multiplatform (KMM), Adaptive & Auto Layout, Storyboards, XIBs, Interface Builder, UI State Preservation, Widgets, Live Activities, CarPlay, Enterprise APIs
- **Machine Learning & AI:** Core ML, Create ML, TensorFlow Lite, ONNX, ML APIs, NLP, Image Classification, Object Detection, Predictive Modeling, Vision APIs
- **Testing & Debugging:** XCTest, XCUITest, TestFlight, Appium, UIAutomation, OCUnt, OCMock, Charles Proxy, Instruments
- (Time Profiler, Allocations, Leaks, UI Debugger), Automated Testing, Unit Testing, UI Testing, TDD
- **Continuous Integration & Deployment:** Jenkins, Hudson, Travis CI, CircleCI, Fastlane, GitHub Actions, Azure Pipelines, Firebase App Distribution
- **Databases & Persistence:** SQLite, MySQL, PostgreSQL, SQL Server, Realm, MongoDB, Cassandra, Firebase Realtime Database, DynamoDB, Core Data
- **APIs & Integration:** RESTful APIs, GraphQL, gRPC, Protobuf, JSON, SOAP, WebSockets, Firebase, AFNetworking, Alamofire, MapKit JS
- **Cloud & Backend Services:** AWS (S3, Lambda, Cognito), Azure (Data, Storage, Pipelines), Google Cloud (Firestore, Pub/Sub), BaaS (Plaid, Synctera, Sardine), Payment Gateways (Stripe, PayPal, Square, Plaid API), PCI DSS Compliance
- **Version Control & Collaboration Tools:** Git, GitHub, GitLab, Bitbucket, SVN, Jira, Confluence, Trello, Slack, Kanban
- **UI/UX & Design Tools:** Sketch, Zeplin, Figma, Adaptive Animations, Apple HIG Compliance
- **Networking & Communication:** Wi-Fi, Bluetooth, NFC, Zigbee, MQTT, LTE, 5G, iBeacon, PTCRB/GCF Standards, Enterprise Mobility
- **Other Tools & Productivity:** Xcode, Visual Studio, Eclipse, Jupyter, Microsoft Office Suite

WORK EXPERIENCE

Walmart, Bentonville, AR

July 2021- Present

Sr. iOS Engineer

Project Domain: Retail & E-Commerce – Subscription Features for Walmart iOS App

App link: <https://apps.apple.com/us/app/walmart-shopping-savings/id338137227>

Team Size Led: 25+ members (6 iOS, 6 Android, 6 Web, 8 Backend, 2 QA, 2 PMs, 1 Scrum Master)

Environment: Swift 5, Xcode 14–16, MVVM-C, Combine, GraphQL, JSON, XCTest, XCUIElement, GitHub, Jenkins, Charles Proxy, Splunk, JIRA

Responsibilities:

- Enhanced Walmart iOS app with subscription features, allowing recurring purchases and automatic deliveries.
- Architected and developed the Subscriptions module using MVVM-C and dependency injection.
- Built the Manage Subscriptions page to update delivery frequency, quantities, and preferences.
- Optimized subscription checkout flow, reducing user clicks and improving usability.
- Created programmatic UI in Swift 5/Xcode with localizable and accessible views.
- Developed GraphQL queries/mutations, consumed JSON APIs, and integrated with Combine for reactive updates. Applied delegation and notification design patterns to handle user inputs and real-time view updates.
- Ensured accessibility compliance using VoiceOver and Accessibility Inspector.
- Maintained 80%+ unit test coverage with XCTest; developed UI automation tests with XCUIElement.
- Debugged APIs using Charles Proxy and resolved production issues with Splunk monitoring.
- Delivered 100+ pull requests and maintained CI/CD pipelines via GitHub and Jenkins.
- Mentored junior developers, assisted with onboarding, and acted as interim iOS team lead on subscription enhancements. Used NSTextAttachment to create labels with inline images, improving UI presentation.
- Participated in sprint planning, backlog grooming, code reviews, and Agile ceremonies.
- Collaborated with QA, product, and design teams to deliver user-friendly, production-ready features.
- Fixed multiple features, analytics, and accessibility defects reported during testing and production.

Macy's, San Francisco, CA

May 2018 – Jun 2021

Lead iOS Mobile Engineer

Project Domain: Retail & E-Commerce – Bag Module Migration & Enhancements for Macy's iOS App

App Link: [Macy's iOS App](#)

Team Size Led: 11 members (PM, Product Manager, 2 iOS, 2 Android, 2 Backend, 3 QA)

Environment: Swift 4, Objective-C, Xcode 9–10, VIPER, CocoaPods, GRDB, Auto Layout, XCTest, Fabric, Jenkins, Git, GitLab, SourceTree, VersionOne, Confluence, Charles Proxy, Agile/Scrum, REST/JSON, Firebase, Agile/Scrum

Responsibilities:

- Migrated the Bag module from Objective-C to Swift 4, improving code maintainability, testability, and runtime performance.
- Architected the Bag feature using VIPER design pattern to enforce separation of concerns and scalability.
- Developed a custom CocoaPod networking framework with Alamofire for REST/JSON API handling, caching, and error recovery.
- Enhanced Bag UI/UX with Auto Layout, stack views, trait collections, and adaptive design for iPhone/iPad universality.
- Integrated Macy's Rewards & Offers APIs into the Bag workflow, enabling promo code persistence, loyalty points tracking, and personalized offers.
- Built an offline-first promo code persistence layer using GRDB (SQLite), ensuring checkout continuity without network access.
- Implemented Apple Pay and WalletKit integration for secure, frictionless in-app payments aligned with PCI DSS standards.
- Added Push Notifications & In-App Messaging for "app-only" offers, loyalty alerts, and price drop notifications via Firebase Cloud Messaging.
- Integrated Barcode & QR Code Scanner using AVFoundation to let users scan in-store items for price, reviews, and availability.
- Applied VoiceOver, Dynamic Type, and Apple HIG compliance to deliver an accessible experience for all users.
- Leveraged delegation, notification center, and Combine-like reactive streams for Bag synchronization across modules.
- Ensured unit and UI test coverage with XCTest/XCUITest and automated builds/test distribution via Fabric and later Firebase App Distribution.
- Optimized API payload parsing and rendering by adopting Codable models and background thread processing.

- Used Charles Proxy and Instruments (Time Profiler, Allocations, Leaks) to debug networking, fix memory leaks, and profile performance bottlenecks.
- Managed CI/CD pipelines in Jenkins and GitLab, integrating automated testing, static analysis, and build distribution.
- Coordinated cross-platform dependencies with Android, Backend, and Payment teams, aligning sprint deliverables and release cadence.
- Acted as interim Scrum lead, handling sprint planning, backlog grooming, and daily standups to keep development on track.

SunTrust, Atlanta, GA

Jul 2017 – May 2018

Senior Mobile Developer

Project Domain: Mobile Banking – Account Management, Payments, Rewards, and Fraud Alert Features

App Link: [SunTrust Mobile App](#) (*App later merged into Truist Financial Corporation*)

Team Size Led: 15+ members (iOS, Android, Backend, QA, PM, Design)

Environment: Swift 3/4, Objective-C, Xcode 8–9, MVVM, Core Graphics, Core Animation, Auto Layout, CocoaPods, TestFlight, GitHub, Jenkins, JIRA, Instruments, Keychain, Agile/Scrum

Responsibilities:

- Audited and refactored the legacy Objective-C codebase, leading the transition to Swift 3/4 for improved performance, readability, and long-term maintainability.
- Re-architected modules using the MVVM design pattern, ensuring separation of concerns and easier testability across critical banking flows.
- Developed new and enhanced existing UI components with Auto Layout, Core Graphics, and Core Animation, modernizing the look and improving responsiveness across iPhone and iPad.
- Implemented CocoaPods for dependency management, standardizing third-party integrations across the mobile team.
- Distributed internal builds through TestFlight, managing provisioning profiles and certificates for QA and UAT testing.
- Strengthened application security by enforcing Keychain-based credential storage, auditing data persistence, and preventing insecure transmission of sensitive information.
- Built unit tests with XCTest and applied TDD practices to achieve higher code coverage and reduce production regressions.
- Integrated the project with GitHub and Jenkins, enabling automated CI pipelines and faster, more reliable delivery cycles.
- Diagnosed and resolved memory leaks and performance bottlenecks using Xcode Instruments (Allocations, Time Profiler, Leaks).
- Collaborated with product managers, backend engineers, and QA teams to deliver account management, payment processing, and fraud alert features.
- Led sprint planning, backlog grooming, and Agile ceremonies, ensuring smooth delivery of mobile banking enhancements.
- Documented application workflows, features, and test cases to support knowledge transfer across engineering teams.
- Served as a technical liaison with business stakeholders, providing progress updates, aligning priorities, and preparing the app for App Store releases.
- Integrated Touch ID/Face ID authentication using LocalAuthentication framework, enhancing secure logins and reducing password dependency.
- Implemented real-time transaction and fraud alerts with Apple Push Notification Service (APNs), ensuring reliable delivery and user engagement.
- Optimized network layer performance by migrating legacy synchronous API calls to asynchronous RESTful services using NSURLSession/Alamofire.
- Configured Splunk and Crashlytics (Fabric) for production monitoring, crash reporting, and proactive debugging of live mobile issues.

TD Ameritrade, Omaha, NE

Jan 2016 – Jun 2017

Senior iOS App Developer

Project Domain: Mobile Trading & Banking – Account Access, Payments, Research, and Trade Execution

App Link: [TD Ameritrade Mobile](#) (Now merged with The Charles Schwab Corporation)

Team Size Led: 12+ members (iOS, Backend, QA, PM, UX)

Environment: Swift 1.2/2.x, Objective-C, Xcode 6–7, MVVM, Cocoa Touch, WatchKit, AVFoundation, Auto Layout, Dependency Injection, Instruments, TestFlight, SiteCatalyst, Localytics, Keychain, Agile/Scrum

Responsibilities:

- Participated from project inception, contributing to app design, architecture, and feature planning, while also driving implementation and deployment.
- Refactored large portions of the legacy Objective-C codebase and transitioned key modules to Swift 2, improving maintainability and reducing technical debt.
- Designed custom UI components with IBDesignables and modernized multiple flows for better usability while adhering to Apple Human Interface Guidelines (HIG).
- Architected core modules with MVVM and dependency injection, enabling testable, decoupled, and scalable trading features.
- Added support for Apple Watch (WatchKit), extending account notifications and trade alerts to wearable devices.
- Integrated AVFoundation to support smooth media playback and in-app video streaming for research and educational content.
- Implemented SiteCatalyst and Localytics for advanced user behavior tracking and analytics-driven product decisions.
- Improved performance by profiling with Instruments (Leaks, Allocations, Time Profiler), resolving memory leaks, and optimizing view controllers.
- Strengthened app security with Keychain Services for secure credential storage and session handling.
- Delivered localized versions of the app, enabling multi-language and region-specific support for global investors.
- Built and distributed QA and UAT builds via TestFlight, supporting iterative feedback cycles.
- Collaborated with cross-functional teams (product, design, QA, backend) to deliver account access, money transfers, check deposit, and trading workflows.
- Provided technical consultancy and team mentorship, improving workflows for source control, CI/CD, and Agile task management.
- Acted as a liaison with business stakeholders, communicating project status, risk, and App Store approval readiness.
- Implemented remote push notifications and background fetch APIs to deliver timely trade confirmations, balance alerts, and market updates without requiring manual refresh.
- Designed offline-ready modules using NSCache and Core Data, enabling users to access account balances, watchlists, and recent trades even during poor network conditions.
- Built Today Widget extensions to allow investors to quickly glance at portfolio performance and stock prices directly from the iOS Notification Center.
- Collaborated with compliance teams to align app features with FINRA/SEC guidelines, while ensuring WCAG accessibility standards with VoiceOver, Dynamic Type, and proper semantic labels.

Mastercard, Purchase, NY
iOS Mobile Developer

Jun 2015 – Jan 2016

Project Domain: Financial Services & Mobile Payments – Mastercard Nearby ATM & Merchant Finder

App Link: [Mastercard Nearby](#)

Team Size Led: 8+ members (iOS, Server, QA, UX, PM)

Environment: Objective-C, early Swift POCs, Xcode 5–6, Cocoa Touch, CocoaPods, MapKit, Core Location, Core Animation, JSON, AFNetworking, Unit Testing, Agile/Scrum

Responsibilities:

- Implemented core features of Mastercard Nearby app, enabling users to locate 2M+ ATMs worldwide and participating merchants in 200+ countries.
- Developed app primarily in Objective-C and created POCs in Swift 1.0 during transition phase to assess feasibility for future projects.
- Integrated RESTful APIs for ATM and merchant data, using AFNetworking for network communication and JSON parsing for structured responses.
- Enhanced location services with MapKit and Core Location, providing step-by-step navigation and country-specific ATM/merchant searches.
- Implemented sophisticated search filters (drive-through, 24-hour, EMV, Maestro acceptance) for precise ATM discovery.
- Applied Core Animation to improve map-based interactions and deliver a more intuitive UI/UX experience.
- Built reusable framework components for location and merchant data, standardizing modules for reuse across Mastercard's iOS apps.
- Collaborated with server team to integrate secure authentication and ensure high availability of financial services data.
- Conducted unit testing and peer code reviews to ensure quality, performance, and adherence to App Store compliance guidelines.
- Participated in app submission process, ensuring conformity to Apple HIG and payment industry security requirements.
- Used Charles Proxy and Xcode Instruments (Leaks, Allocations, Time Profiler) to profile app, debug issues, and optimize performance.

- Coordinated with QA and product teams for defect triaging, sprint deliverables, and iterative feature enhancements.
- Collaborated cross-functionally to support refinements and post-release enhancements based on user adoption and analytics.

Frontier Airlines, Denver, CO

Jan 2014 – Jun 2015

iOS Mobile Developer

Project Domain: Airline & Travel – Flight booking, seat upgrades, bundles, & loyalty program management for the Frontier app

App Link: [Frontier Airlines](#)

Team Size Led: 8 members (2 iOS, 2 Android, 2 Backend, 1 QA, 1 PM)

Environment: Objective-C, Xcode 4.6/5, MVC, Cocoa Touch, CocoaPods, AFNetworking, Storyboards, Auto Layout, XCTest, Git, JIRA, Agile

Responsibilities:

- Developed booking, check-in, and trip management features in Objective-C following MVC architecture.
- Consumed RESTful APIs with AFNetworking, handling JSON payloads for flights, baggage, and loyalty miles.
- Built UI with Storyboards and Auto Layout, ensuring adaptive layouts for both portrait and landscape.
- Integrated third-party libraries via CocoaPods to modularize dependencies and speed up development.
- Optimized workflows for seat upgrades, bundles, and add-on purchases, improving usability.
- Designed custom views per iOS 7 HIG, aligning app visuals with Apple's flat design transition.
- Managed backlog and user stories in JIRA; actively participated in Agile ceremonies.
- Implemented XCTest unit tests to validate booking and payment modules before App Store submission.
- Used Git extensively for source control and code reviews with the iOS team.
- Debugged API issues with Charles Proxy and ensured smooth release cycles.

GS Marketing, Inc., Houston, TX

Mar 2012 – Dec 2012

Software Developer I

Project Domain: Marketing & IT Services – ETL automation, website maintenance, & app development for internal BI systems

Environment: Java, C++, C#, Scala, PHP, HTML, CSS, JavaScript, MySQL, Windows Task Scheduler, Raspberry Pi

Responsibilities:

- Developed ETL processes in C# to anonymize personally identifiable information (PII) in internal databases.
- Implemented applications in Java and C++, including an Address Book system and payment processing module.
- Automated recurring workflows and monthly report generation using Windows Task Scheduler.
- Built and maintained the company website with PHP, HTML, CSS, JavaScript, and MySQL.
- Created algorithms and sequence logic to support data-driven reporting and analytics.
- Debugged and optimized scripts, fixing errors in existing workflows.
- Authored setup documentation for Scala development environments to assist other developers.
- Worked on hardware integration projects with Java/C++ and Raspberry Pi.
- Applied software engineering principles such as Use Case Diagrams, Sequence Diagrams, Life Cycle Models, and Critical Path Analysis.
- Mentored junior developers by providing coding support and documentation guidance.

EDUCATION

Bachelor of Science (B.S.) in Computer Science - *University of Houston, Houston, TX*

(Concentration Software Design, Minor in Mathematics)